

X-Ray Diffraction of Crystals

PC2193: Experimental Physics

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Abstract

This experiment uses X-ray diffraction (XRD) to explore LiF and KBr crystal characteristics and determine Planck's constant h . In Experiment 1, narrow-scan diffractometry of a Lithium Fluoride (LiF) crystal is used to determine its interplanar spacing from four Bragg peaks corresponding to the Cu K_α and K_β characteristic lines at two diffraction orders. In Experiment 2, the Bremsstrahlung (white) spectrum produced at three accelerating voltages (25 kV, 30 kV, 35 kV) is analysed to determine the short-wavelength cutoff angle, from which Planck's constant h is derived via the Duane–Hunt relation. In Experiment 3, the same diffractometric procedure is repeated for a Potassium Bromide (KBr) crystal, which produces six observable Bragg peaks across three diffraction orders. Results are compared with accepted literature values. The experimental interplanar spacing for LiF was found to be $2.0053 \pm 0.0037 \text{ \AA}$ (literature: $2.008 \text{ \AA}$), for KBr $3.2756 \pm 0.0165 \text{ \AA}$ (literature: $3.290 \text{ \AA}$), and Planck's constant was determined to be $6.5151 \pm 0.1328 \times 10^{-34} \text{ J} \cdot \text{s}$ (literature: $6.626 \times 10^{-34} \text{ J} \cdot \text{s}$).

1 Background & Objectives

1.1 The Underlying Physics

X-ray diffraction analysis is a technique that exploits the wave nature of X-rays to probe the periodic atomic structure of crystalline materials. Because the wavelengths of X-rays ($0.1\text{--}10 \text{ \AA}$) are comparable to the spacing between atomic planes in a crystal, the crystal acts as a three-dimensional diffraction grating, producing interference patterns that carry structural information.

1.1.1 X-Ray Generation

When electrons emitted from a heated cathode are accelerated through a high voltage V and strike the copper anode, two types of X-ray radiation are produced.

Bremsstrahlung (continuous radiation). As an electron decelerates in the Coulomb field of a target nucleus, it radiates a photon. Because this energy loss can be any fraction of the electron's initial kinetic energy eV , the resulting photon spectrum is continuous. The upper energy limit — or equivalently the **minimum wavelength** $\lambda_{\min}$ — corresponds to the case in which the electron surrenders its entire kinetic energy in a single collision:

$$eV = \frac{hc}{\lambda_{\min}} \quad (1)$$

Characteristic radiation. If the bombarding electron possesses sufficient energy to eject a core (K-shell) electron from a copper atom, the resulting vacancy is filled by an electron from an outer shell, releasing a photon whose energy equals the inter-shell energy difference. For the copper anode, this produces two sharp peaks: the K_α line ($L \rightarrow K$ transition, $\lambda = 1.5406 \text{ \AA}$) and the K_β line ($M \rightarrow K$ transition, $\lambda = 1.3922 \text{ \AA}$).

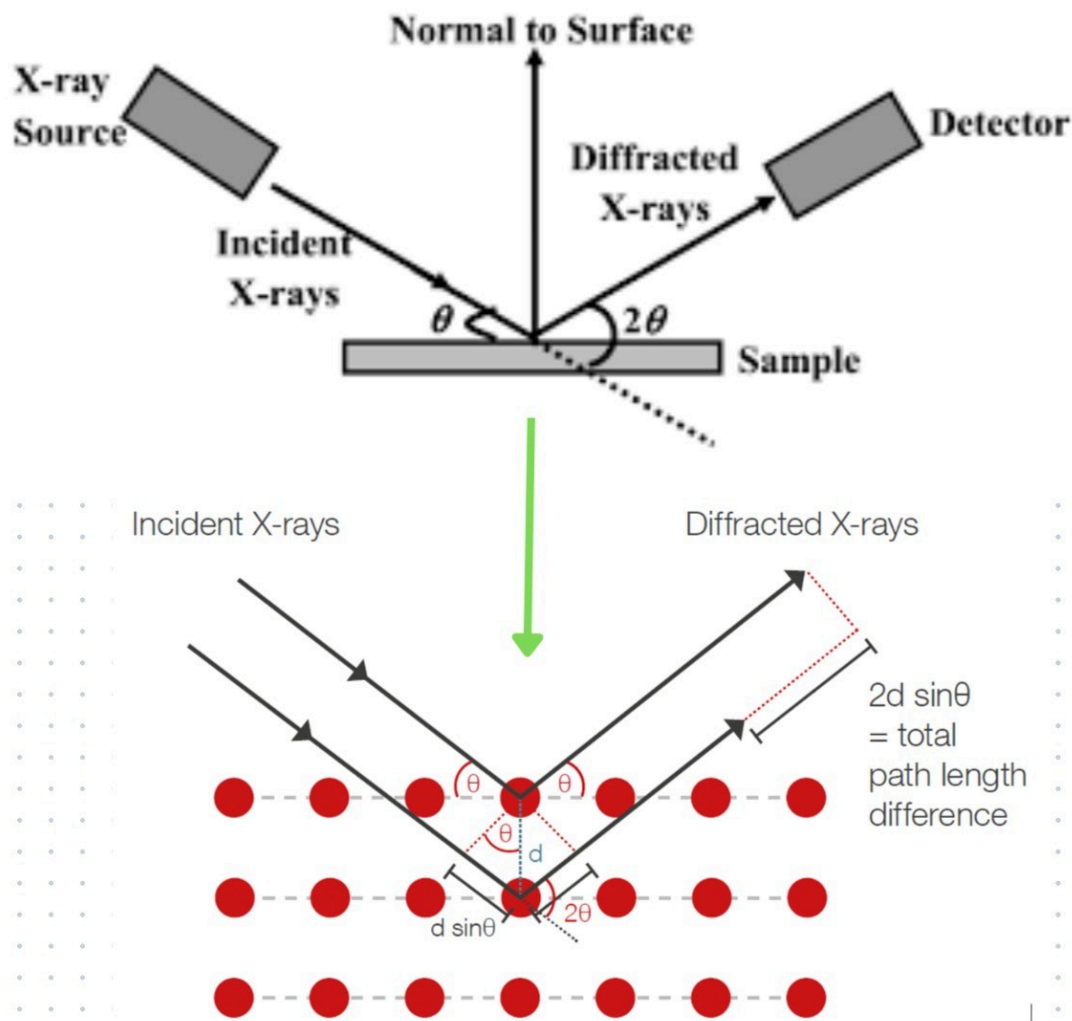


Figure 1: Schematic of the path of the X-rays after emission and how the path difference leading to diffraction arises. The green arrow points to how the X-rays reflect and diffract off of the sample. The red dots represent atoms organised in the crystal lattice.

1.1.2 Bragg Diffraction

When an X-ray beam of wavelength λ impinges on a crystal at a glancing angle θ with respect to a set of parallel atomic planes separated by spacing d , waves reflected from successive planes travel different path lengths. Constructive interference corresponds to a sharp diffraction peak and occurs when this path difference equals an integer multiple of the wavelength. As shown in Figure 1, the path difference between reflections from adjacent planes is $2d \sin \theta$, giving **Bragg's law**:

$$n\lambda = 2d \sin \theta \quad (2)$$

where $n = 1, 2, 3, \dots$ is the order of diffraction. In the PHYWE goniometer, the crystal rotates by angle θ while the detector arm moves at twice the rate (2:1 coupling), so all measured angles are the **detector angle**.

1.2 Objectives

1. Use XRD to determine the interplanar spacing of LiF and compare with the accepted value.
2. Measure Planck's constant h via the short-wavelength limit of the Bremsstrahlung spectrum.
3. Use XRD to determine the interplanar spacing of KBr and confirm the identity of the crystal.
4. Evaluate discrepancies between experimental and literature values and account for key sources of uncertainty.

2 Experimental Setup and Procedure

2.1 Apparatus

The key components are listed in Table 1 and the full apparatus is shown in Figure 2.

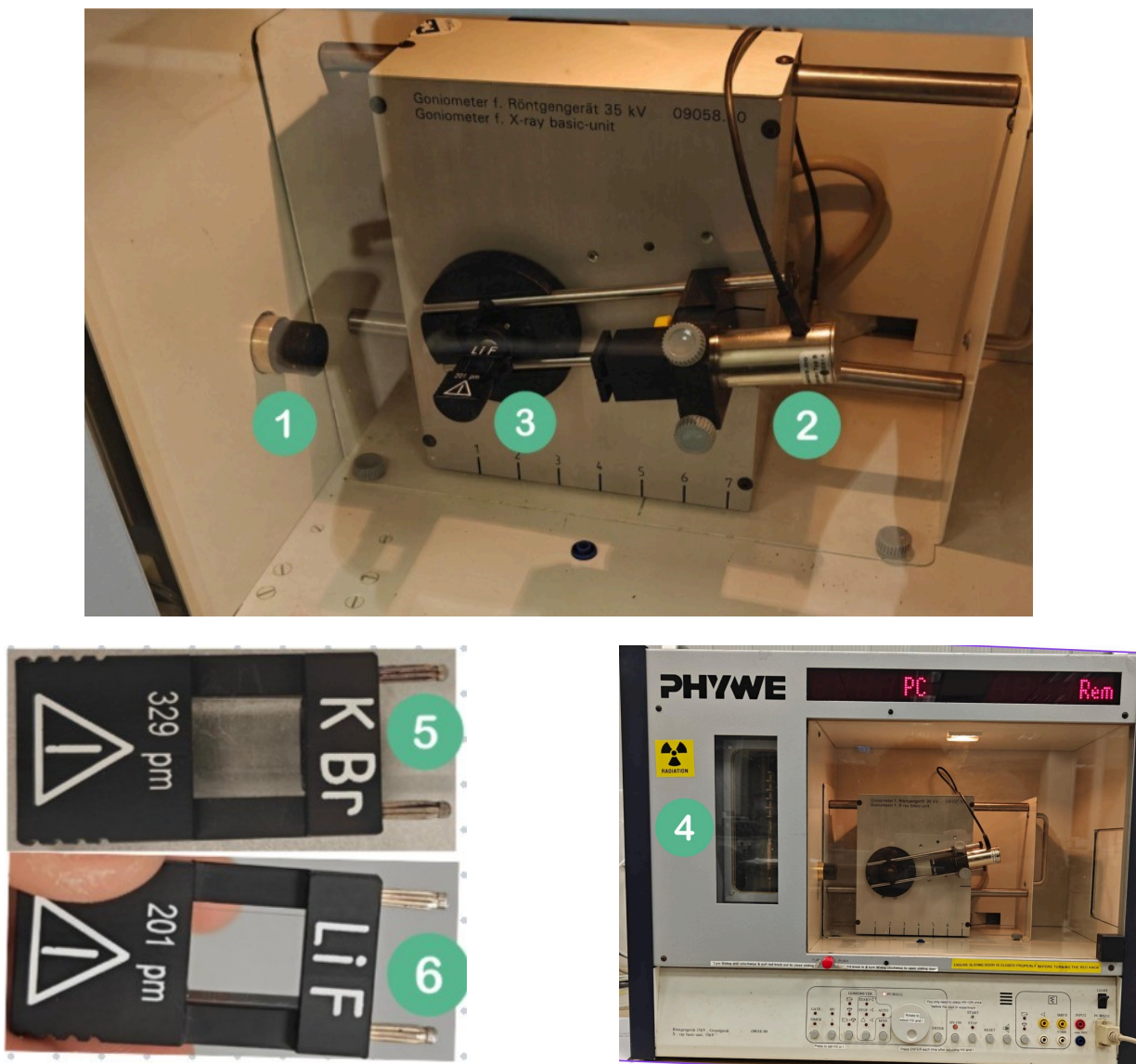


Figure 2: (Top) Main schematic of the XRD setup. (Bottom Left) Close-up of the LiF and KBr crystals. (Bottom Right) The whole setup.

Number	Component	Description
1	Collimator	Directs the X-rays from the source into a defined beam towards the sample
2	Geiger–Müller detector	Counts individual X-ray photons; outputs integer count rates per second to the MEASURE software.
3	Sample in the sample holder	(2) and (3) are part of the goniometer assembly. The goniometer is a stepper-motor driven, 2:1 coupled mode (detector moves at twice the crystal angle). Bragg–Brentano geometry.
4	Copper X-ray tube	Heated cathode; copper anode ($Z = 29$); operating voltage 0–35 kV; tube current 0–1.0 mA. Emits Cu K_α (1.5406 Å) and K_β (1.3922 Å) characteristic lines superimposed on the Bremsstrahlung continuum.
5	KBr crystal	Potassium bromide; rock-salt structure; accepted interplanar spacing $d = 3.290$ Å
6	LiF crystal	Lithium fluoride; rock-salt structure; accepted interplanar spacing $d = 2.008$ Å.

Table 1: Principal components of the experimental setup

2.2 Procedure

All parameters were set through the MEASURE software. The instrument control panel was not used.

General procedure was followed from the lab manual. Only the notable modifications and choices of number of runs have been listed here.

2.2.1 Experiment 1: Lattice Constant of LiF

Two wide-scans were conducted with 5 narrow scans for each peak (thus, total 40 narrow scans).

- Voltage: 35 kV; Current: 0.05 mA
- Scan range: $\theta = 5^\circ$ to 55° ; step: 0.5° ; integration: 5 s per step. For 2nd wide-scan, step was kept at 0.2° and integration time at 5 s.

2.2.2 Experiment 2: Planck’s Constant

- Variable Voltage mode: 25 kV, 30 kV, 35 kV
- Scan range: $\theta = 5^\circ$ to 20° ; step: 0.1° ; integration: 1 s
- The scan was repeated **7 times**. The last scan was conducted at a higher current to see if the noise reduced.

The averaged count-rate curves were then analysed to find $\theta_{\min}$ for each voltage by **linear extrapolation** of the rising edge of the Bremsstrahlung onset back to the background level. The background was estimated by considering mean impulse counts before the Bremsstrahlung arose.

2.2.3 Experiment 3: Lattice Constant of KBr

The LiF crystal was replaced with the KBr crystal. Identical procedure (with listed modifications) from Experiment 1 was followed.

- 1st wide scan and corresponding narrow scans conducted at 0.5 mA current. 2nd conducted at 0.4 mA. This was done for better visibility of the 5th peak.
- Total of 60 narrow scans.

3 Results and Analysis

3.1 Raw Data

The graphs generated by the measure software immediately upon the completion of a measurement were taken as the raw-data and also to confirm the presence of peaks (for exp 1 and exp 3) and the expected Bremsstrahlung for exp 2.

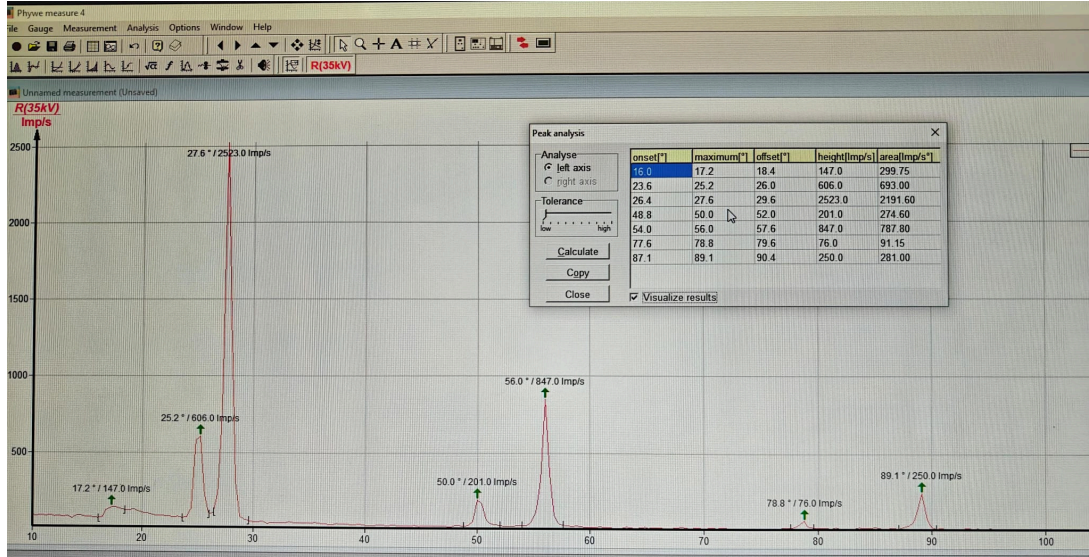


Figure 3: Graph generated by MEASURE after the completion of the widescan of KBr

For exp 1 and exp 3 the inbuilt **peak analysis** function of the MEASURE software (as visible in figure 3) was used to identify the θ ranges to conduct the narrow scans.

For exp 2, the graphs generated by MEASURE were used to confirm the Bremsstrahlung and check for the noise and separation between the 25 kV, 30 kV and 35 kV curves.

3.2 Error Analysis

- Error in θ for exp 1 and 3 is taken as the standard deviation of θ_{peak}
- Error for θ_{min} in exp 2 is determined by considering the fit error.
- Error for known values and constants such as e , λ for K_α and K_β , c has been taken to be negligible.
- Since equations for the Plancks constant (1) and the inter-lattice spacing d (2) both involve trigonometric ratios, the error is propagated according to the general error propagation formula in quadrature.

If,

$$Q = f(x_1, x_2) \quad (3)$$

Then,

$$\frac{\Delta Q}{Q} = \sqrt{\left(\left(\frac{\partial f}{\partial x_1}\right) \frac{\Delta x_1}{x_1}\right)^2 + \left(\left(\frac{\partial f}{\partial x_2}\right) \frac{\Delta x_2}{x_2}\right)^2} \quad (4)$$

Where we can also have x_3, x_4 and so on. Thus, for equation (2) involving a $\sin \theta$ term, the error of d also depends on $\frac{d}{d\theta} \left(\frac{1}{\sin \theta} \right) = |\operatorname{cosec} \theta \cot \theta|$. This simplifies to just $\cot \theta$.

- The detailed calculations for the errors have been omitted for the sake of simplicity but they are available on my laptop.

3.3 Analysis of Data

3.3.1 Experiment 1: Lattice Constant of LiF

For each of the four peaks, the weighted-average peak centre was computed for each of the five trial narrow scans. The following formula was used:

$$\overline{2\theta}_{\text{peak}} = \frac{\sum_i (2\theta)_i \cdot R_i}{\sum_i R_i} \quad (5)$$

Where R_i is the impulse count at that value of 2θ .

The mean $\langle 2\theta \rangle$ was calculated across the ten (round 1 + round 2) trials. The interplanar spacing d was then determined from equation (2) through each peak and then the mean was taken to reach the final experimental value.

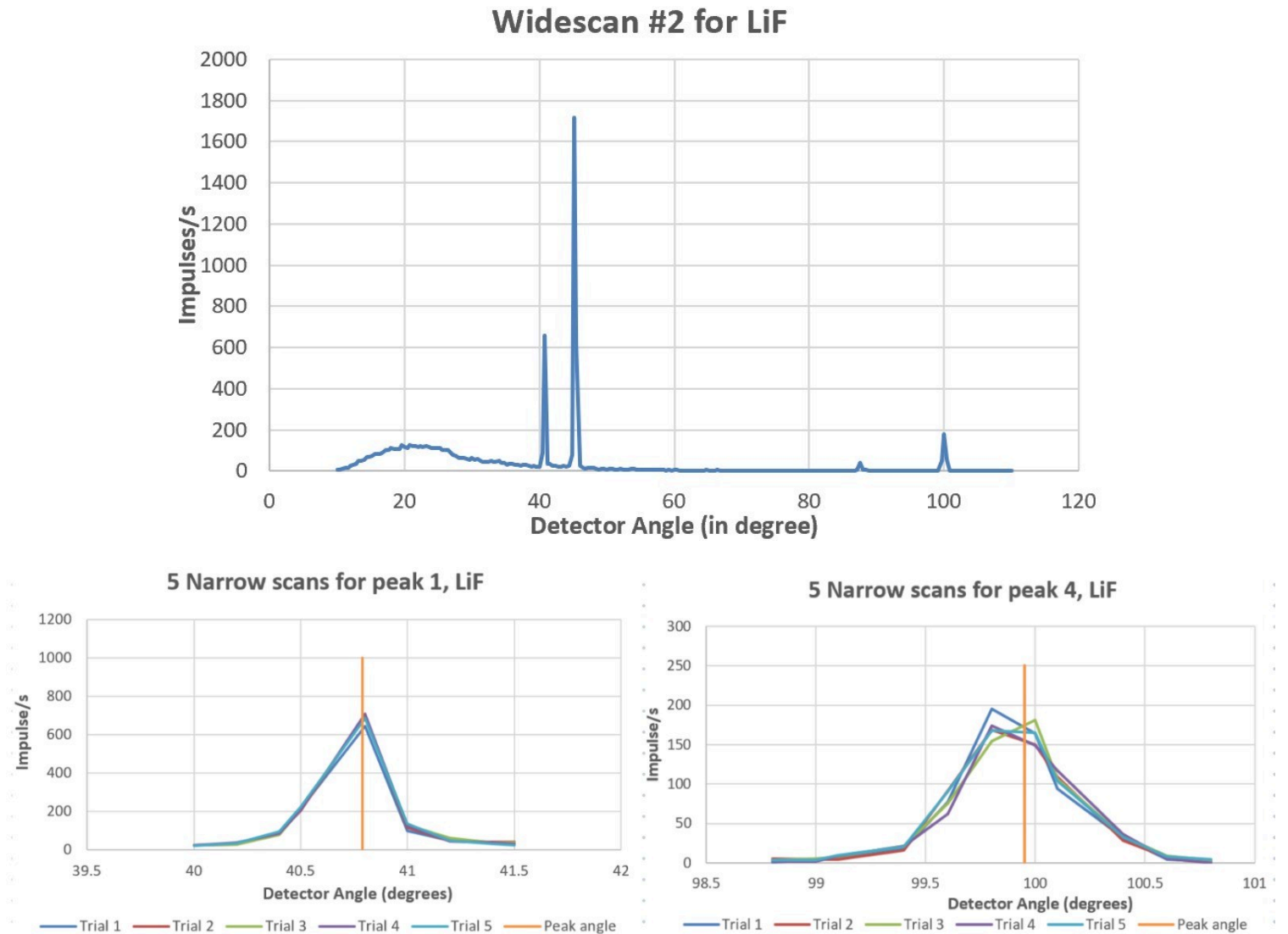


Figure 4: (Upper) Widescan for LiF at 0.2 degree angle step. (Bottom left) 5/10 narrow scans for peak 1 with peak angle. (Bottom right) 5/10 Narrow scans for peak 4 with peak angle.

Peak	Assignment	$\langle 2\theta \rangle$ (°)	$\delta_{2\theta}$ (°)	d_{exp} (Å)
1	$n = 1, K_{\beta}$	40.7895	0.0291	1.9975
2	$n = 1, K_{\alpha}$	45.3011	0.0752	2.0002
3	$n = 2, K_{\beta}$	87.5896	0.0795	2.0116
4	$n = 2, K_{\alpha}$	99.9614	0.0587	2.0118
Mean				2.0053

Table 2: Bragg peak analysis for LiF. Peak centres from the weighted-average method, interplanar spacings d_{exp} calculated via Bragg's Law. Literature value: $d_{\text{lit}} = 2.008 \text{ Å}$

Experimentally determined value = **$2.0053 \pm 0.0037 \text{ Å}$**

3.3.2 Experiment 2: Planck's Constant

For each of the three accelerating voltages, the rising edge of the averaged Bremsstrahlung spectrum was identified (see Annex for method), a linear fit was applied, and the fit was extrapolated to the background level to obtain $(2\theta)_{\text{min}}$. Table 3 summarises the values of θ_{min} and the resulting values of h .

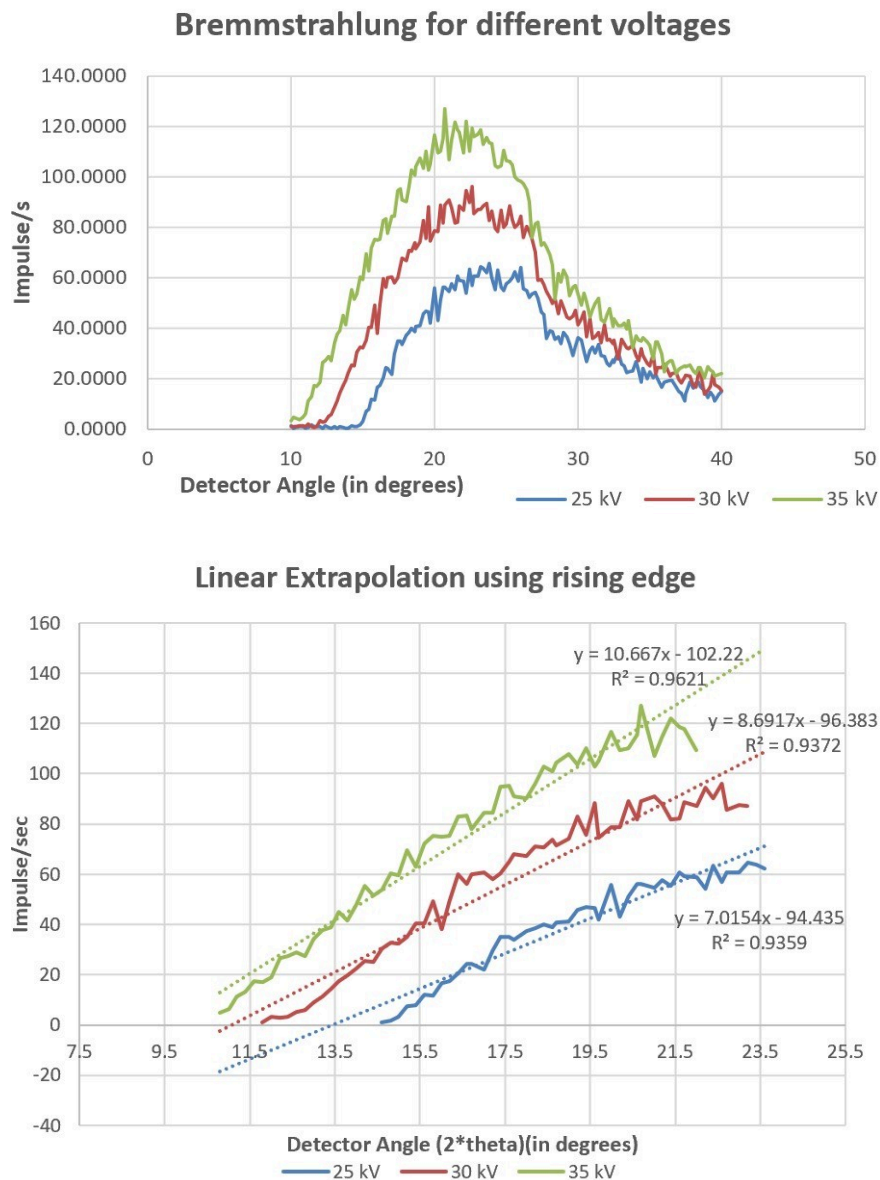


Figure 5: (Upper) Onset of the Bremmstrahlung graphed. (Lower) Linear extrapolation using selected values.

In figure 5 (lower), we can see the equations of the lines of linear regression. These equations have been used to determine the $(2\theta)_{\min}$ value by setting $y = \text{background level}$.

Voltage	$(2\theta)_{\min}$ (°)	$\delta(2\theta)_{\min}$ (°)	$\lambda_{\min}$ (Å)	h_{exp} (J·s) (10^{-34})	δh (J·s) (10^{-34})	% discrepancy
25 kV	13.5798	0.494	0.4741	6.3345	0.2288	4.39
30 kV	11.2382	0.332	0.3926	6.2953	0.1851	4.98
35 kV	9.9656	0.204	0.3483	6.5151	0.1328	1.67
Mean				6.3817		

Table 3: Extrapolated cutoff angles, minimum wavelengths, and derived values of Planck's constant for the three accelerating voltages. Literature value: $h_{\text{lit}} = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$. $\lambda_{\min}$ computed via Bragg's law (equation 2) with $d_{\text{LiF}} = 2.0053 \text{ Å}$ (using value obtained in experiment 1) and $n = 1$. We choose the 35 kV value for h as our final experimental result. (See annex)

3.3.3 Experiment 3: Lattice Constant of KBr

The same data analysis procedure as Experiment 1 was applied to the six KBr peaks.

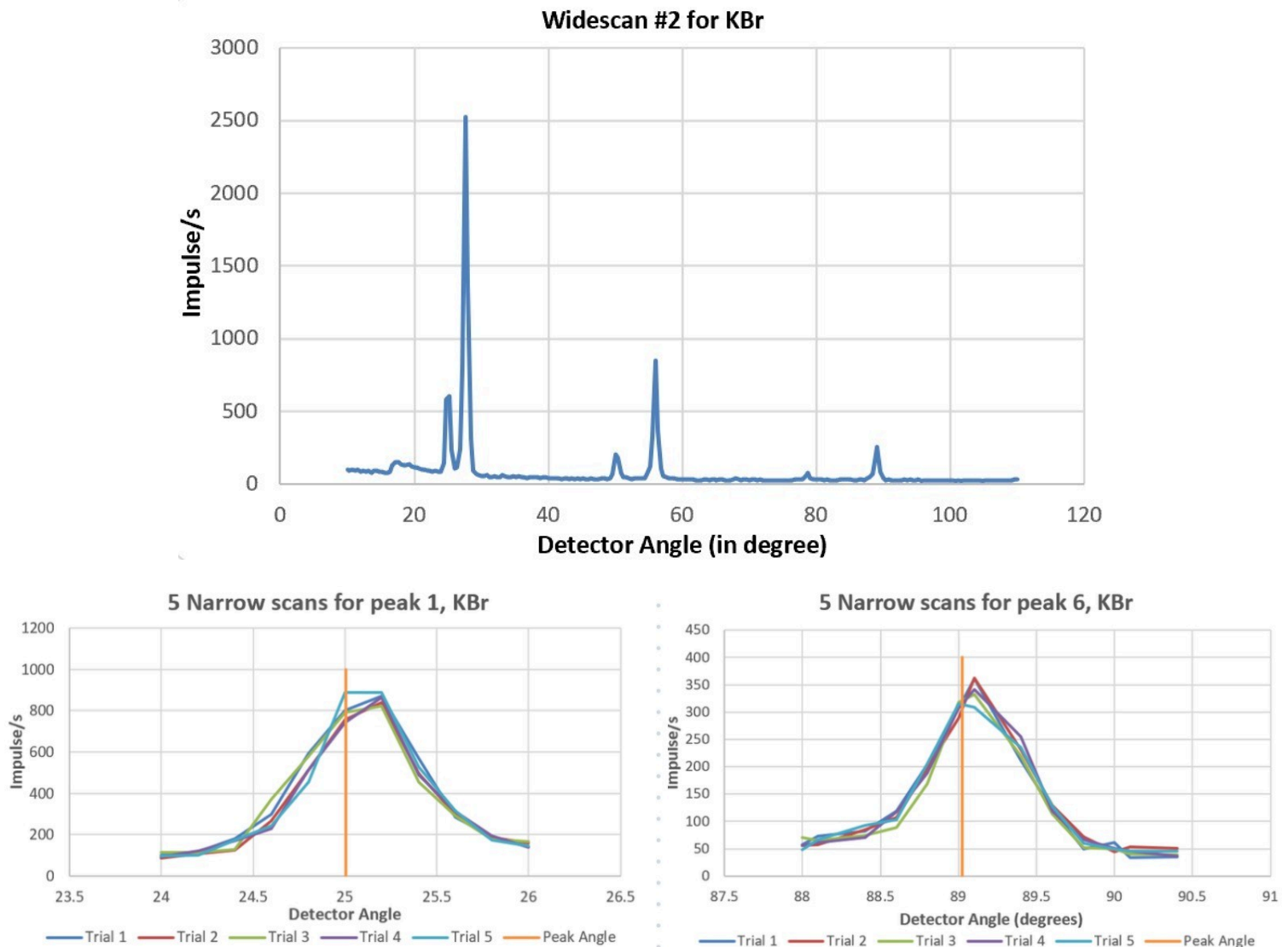


Figure 6: (Upper) Widescan for KBr at 0.2 degree angle step. (Bottom left) 5/10 narrow scans for peak 1 with peak angle. (Bottom right) 5/10 Narrow scans for peak 6 with peak angle.

Peak	Assignment	$\langle 2\theta \rangle$ (°)	$\delta(2\theta)$ (°)	d_{exp} (Å)
1	$n = 1, K_{\beta}$	25.0054	0.0894	3.2154
2	$n = 1, K_{\alpha}$	27.6635	0.0647	3.2220
3	$n = 2, K_{\beta}$	50.2451	0.0502	3.2791
4	$n = 2, K_{\alpha}$	55.8661	0.0125	3.2887
5	$n = 3, K_{\beta}$	78.5665	0.0746	3.2982
6	$n = 3, K_{\alpha}$	89.0219	0.0773	3.2963
Mean				3.2756

Table 4: Bragg peak analysis for KBr across three diffraction orders. Literature value: $d_{\text{lit}} = 3.290$ Å.

Experimentally determined value = **3.2756 ± 0.0165 Å**

4 Discussion

4.1 Summary of Experimental Results

Experiment	Quantity	Experimental	Literature	% Discrepancy
Exp 1 (LiF)	d (Å)	2.0053 ± 0.0037	2.008	0.13%
Exp 3 (KBr)	d (Å)	3.2756 ± 0.0165	3.290	0.44%
Exp 2 (25 kV)	h ($J \cdot s$)	6.3345×10^{-34}	6.626×10^{-34}	4.39%
Exp 2 (30 kV)	h ($J \cdot s$)	6.2953×10^{-34}	6.626×10^{-34}	4.98%
Exp 2 (35 kV) Chosen	h ($J \cdot s$)	$(6.5151 \pm 0.1328) \times 10^{-34}$	6.626×10^{-34}	1.67%
Exp 2 (mean)	h ($J \cdot s$)	6.3817×10^{-34}	6.626×10^{-34}	3.69%

Table 5: Summary of all experimental results compared with accepted literature values.

4.2 Interpretation of Results

4.2.1 LiF and KBr Lattice Constants

The experimentally determined lattice constants for LiF (2.0053 Å) and KBr (3.2756 Å) show strong agreement with literature values, with discrepancies of 0.13% and 0.44% respectively. The consistency of the d values across multiple diffraction orders and both the K_{α} and K_{β} lines reinforces the reliability of the results, as agreement across multiple peaks provides stronger structural evidence than a single peak.

As expected, higher-order peaks yielded slightly more accurate results. This is due to the fact that $\tan \theta$ is larger at higher angles, which minimizes the impact of angular uncertainties on the calculated interplanar spacing. Contrary to the initial expectation in the lab notes, KBr peaks appeared at **smaller** 2θ angles than the corresponding LiF peaks. This is consistent with Bragg's

Law ($n\lambda = 2d \sin \theta$), which dictates that for a constant wavelength, a larger interplanar spacing d must result in a smaller diffraction angle θ .

4.2.2 Planck's Constant

Among the three tested accelerating voltages, the 35 kV trial yielded the most accurate result for Planck's constant (1.67% discrepancy). While lower voltages theoretically provide a larger $\theta_{\min}$ that makes the rising edge more accessible, the 35 kV trial likely benefited from higher X-ray intensity and a better signal-to-noise ratio. All three values were found to be reasonably consistent within their calculated uncertainties. It should also be noted that the experimentally determined value of d is used instead of the literature value, also contributing to the error.

4.3 Leading Sources of Error

The experimental errors are categorized into systematic offsets and random fluctuations.

1. **Zero-point offset of the goniometer:** A misalignment between the nominal $2\theta = 0$ position and the actual beam axis introduces a systematic shift in all measured angles. This error is most pronounced at low angles.
2. **Background radiation and detector noise:** Scattered radiation and detector dark counts create a non-zero baseline below $\theta_{\min}$. This background makes the extrapolated intercept sensitive to how the background level is estimated and the width of the fitting window.
3. **Crystal alignment:** Any slight tilt of the crystal face relative to the beam causes a geometric offset in measured angles that the weighted-average method cannot compensate for.
4. **Integer rounding and Poisson noise:** The Geiger-Müller tube outputs integer counts per second, introducing a rounding error of ± 0.5 Imp/s. Additionally, the 1 s integration time limits the statistical precision, though repeating the scans reduces this effect by $\sqrt{N}$.
5. **Bremsstrahlung onset ambiguity:** Unlike sharp Bragg peaks, the continuum cutoff must be inferred by linear extrapolation. The quality of the fit and the selection of the rising-edge window are the dominant sources of uncertainty in the determination of h .
6. **Takeaway:** Thus, the biggest contributor of uncertainty is the integer recordings of the impulses and short integration times (≤ 5 s). These directly increase error in 2θ measurements thereby propagating error to all the other results.

5 Conclusion

The experimental objectives were successfully achieved. The interplanar spacings were determined to be 2.0053 ± 0.0037 Å for LiF and 3.2756 ± 0.0165 Å for KBr, showing discrepancies of 0.13% and 0.44% from literature respectively. Physical confirmation of the relationship between lattice spacing and diffraction angle was observed, as the larger KBr lattice spacing resulted in peaks at smaller angles than those of LiF.

The 35 kV voltage was chosen as the best estimate ($(6.5151 \pm 0.1328) \times 10^{-34} \text{ J} \cdot \text{s}$) due to its superior signal-to-noise ratio and lowest percentage discrepancy (1.67%). As predicted, the lattice constants were measured with higher accuracy than Planck's constant, because localized Bragg peaks allow for more precise measurement than the extrapolated Bremsstrahlung onset.

Future accuracy could be improved by using higher-order peaks, increasing integration times to reduce Poisson noise. For experiment 2 particularly, more readings can be taken at intermediate voltages and h can be determined through more robust linear regression with more data points.

6 References

- [1] PC2193 Teaching Team. (2025). *X-Ray Diffraction of Crystals* (Lab Manual, Rev. Wk 4). National University of Singapore.
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- [3] PHYWE Systeme GmbH & Co. KG. (2024). *XR 4.0 Expert Unit, 35 kV (09058.99)* — *Product Description*. Retrieved from <https://www.phywe.com>
- [4] Duane, W., & Hunt, F. L. (1915). On X-ray wave-lengths. *Physical Review*, 6(2), 166–172. <https://doi.org/10.1103/PhysRev.6.166>
- [5] Reed, B. C. (2013). Extrapolation of the Bremsstrahlung continuum to determine Planck's constant. *European Journal of Physics*, 34(2), 325. <https://doi.org/10.1088/0143-0807/34/2/325>

7 AI Usage Declaration

I declare that I have used AI in the preparation of this assignment. Claude (Anthropic) was used to: generate the Typst template and formatting structure for this report. Claude was also used to help compile all the data files into organised excel sheets. In general, AI was used to help with troubleshooting when typst code failed to compile or when excel ran into problems.

8 Annex

- For experiment 1 and 3, results from taking the mean of all impulse counts and then finding θ_{peak} vs finding θ_{peak} for all trials and then taking the mean were compared. No significant difference was found. For example, value of d obtained using first method for Exp 1, peak 4 was = 2.011 Angstrom whereas the value of d calculated using second method was also = 2.011 Angstrom. This report uses the latter method (finding peak for all trials and then taking mean)
- **Identification of Bremsstrahlung onset:** The data from the first 6 runs was first averaged (The 7th run was excluded since it was conducted at a higher current so it would artificially inflate the slope if included in the average). A 7 data point moving average was then calculated to reduce the noise in the data. On the basis of this smoothed data, the onset and peak points of the Bremsstrahlung were chosen. The corresponding raw data values were used to conduct the linear extrapolation.

Smoothed Brehmmstrahlung

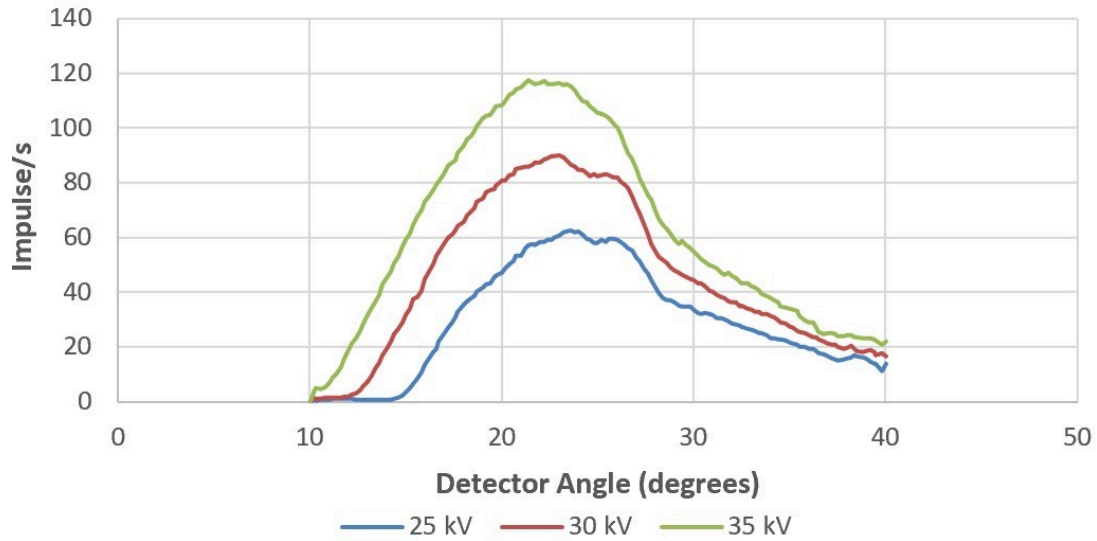


Figure 7: Brehmmstrahlung curves graphed with the smoothed values

The smoothed values made it easier to determine the onset and finish of the linear region.

- **Choice of result for experiment 2** Instead of considering the mean of h -values or a singular reading, we can graph λ_{min} vs $\frac{1}{V}$ and get value of h through the slope.

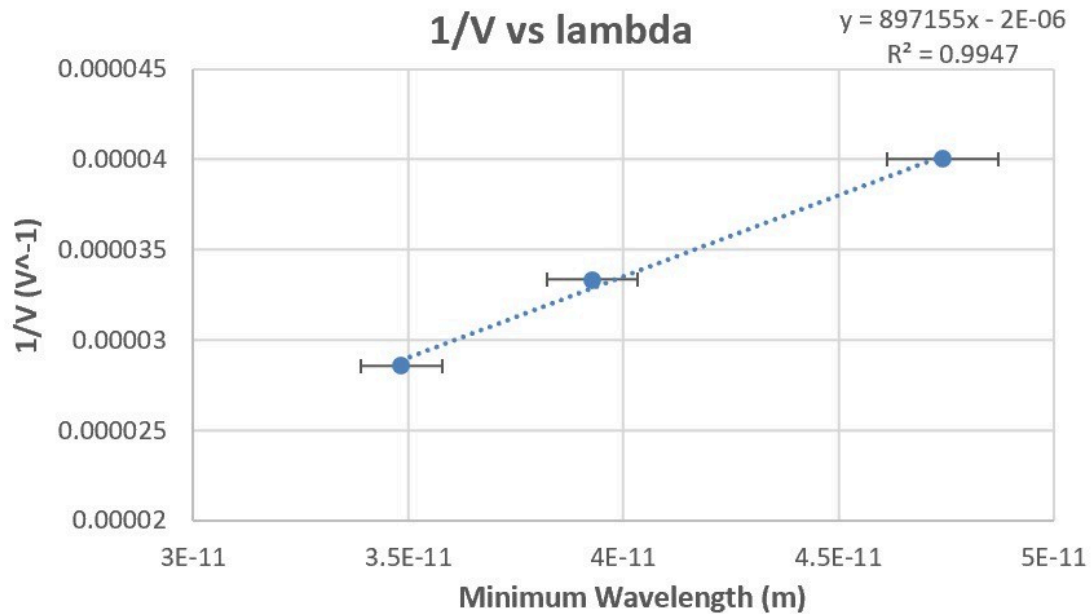


Figure 8: $\frac{1}{V}$ vs λ_{min} with error bars

Here the slope is equal to $\frac{e}{hc}$. We can determine h using this value.

$$h = (5.956 \pm 0.1608) \times 10^{-34} \quad (6)$$

This value is much worse than other previously determined values hence we chose the 35 kV value as our experimental result.